

Actuator-Sensor-Placement in Dynamic Systems

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Preparation

The mode shapes of the cantilever beam were calculated using the following equation:

$$Y_r(x) = A \left(\sin(\beta_r x) - \sinh(\beta_r x) - \frac{\sin(\beta_r L) + \sinh(\beta_r L)}{\cos(\beta_r L) + \cosh(\beta_r L)} (\cos(\beta_r x) - \cosh(\beta_r x)) \right). \quad (1.1)$$

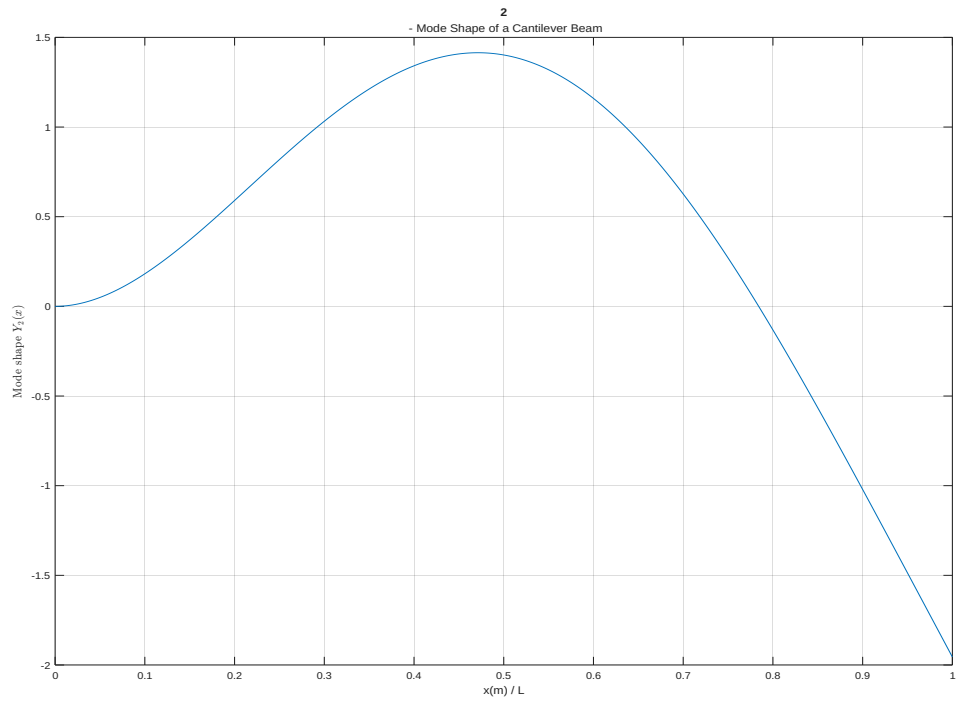
Using $A = 1$ and

$$\begin{aligned} \beta_2 L &= 4.69, \\ \beta_3 L &= 7.85, \\ \beta_4 L &= 10.995, \end{aligned}$$

we plotted the first four mode shapes using MATLAB. The numerically determined node positions (zeros) for modes 2 to 4 are given in table 1.1. The mode shapes are plotted in figure 1 to 1.2.

Mode	1st Node	2nd Node	3rd Node
2	0.7841	-	-
3	0.5039	0.8682	-
4	0.3584	0.6441	0.9056

Table 1.1: Normalized node positions of cantilever beam mode shapes.



captionSecond mode shape.

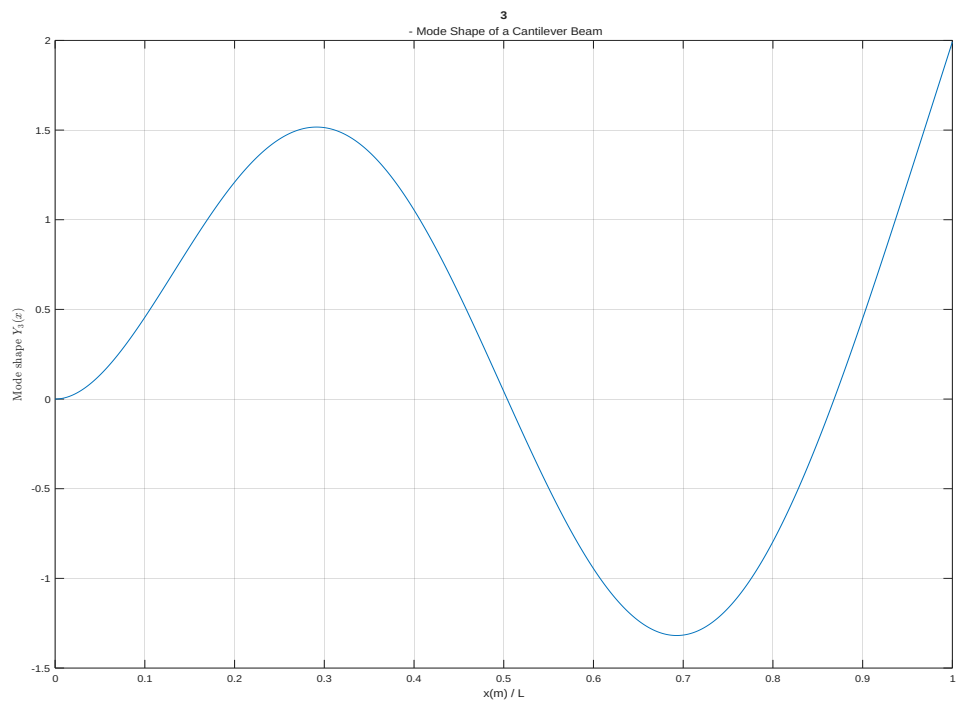


Figure 1.1: Third mode shape.

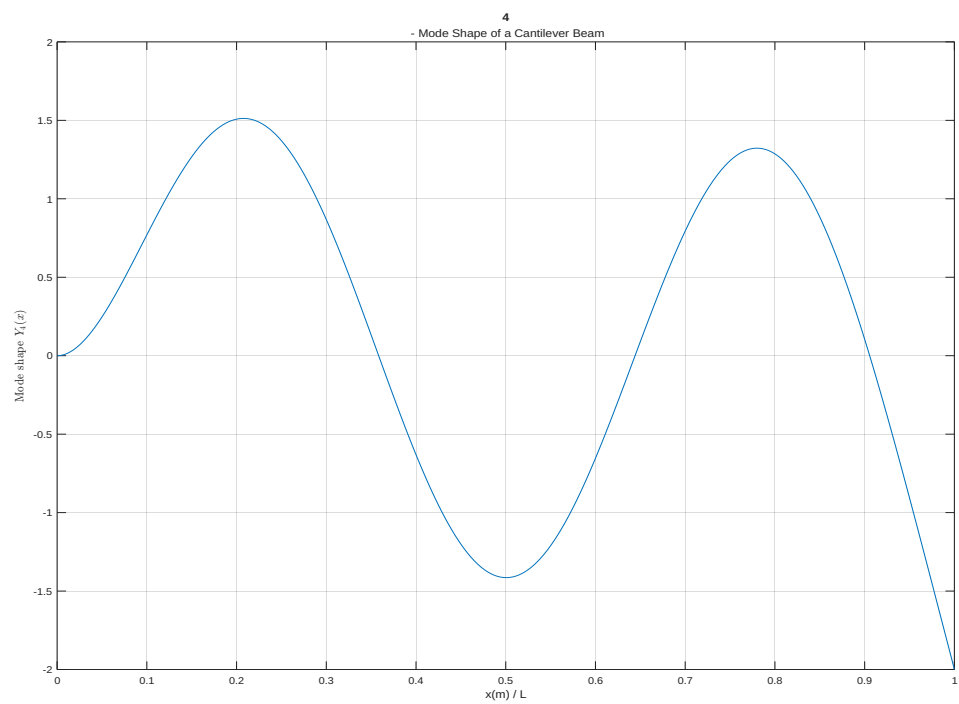


Figure 1.2: Fourth mode shape.

Experiments

2.1 Determination of Resonance Frequencies

The actuator was placed a few centimeters above the bottom end of the cantilever beam, and the accelerometer was placed at the top. The beam was excited using the waveform generator. By manually adjusting the frequency, we identified the resonance peaks:

$$f_2 = 12 \text{ Hz} \tag{2.1}$$

$$f_3 = 36 \text{ Hz} \tag{2.2}$$

2.2 System Response

With the accelerometer mounted at the top of the cantilever beam, we measured the frequency response from 1 Hz to 100 Hz. The bode plot is given in Figure 2.1. Under the assumption that all poles and zeros are on the imaginary axis, we can estimate a pole-zero-map. We had already identified four modes, each corresponding to a resonance frequency.. This also tracks with the Bode plot in Figure 2.1. All the resonances go with a 180 deg phase drop, indicating a pair of complex conjugate poles. We therefore place a pole in the pole-zero-map at

$$s = \pm j2\pi \cdot f_r. \tag{2.3}$$

This results in the pole-zero map in Figure 2.2.

2.3 System Response — Sensor on a Node

From (2.2), we know the frequency of the third mode. By applying this frequency with our actuator and placing the sensor at one of the nodes of this mode shape, the mode should not be observable. Concretely, we should not observe a resonance peak at this frequency in the Bode plot. (e.g., no

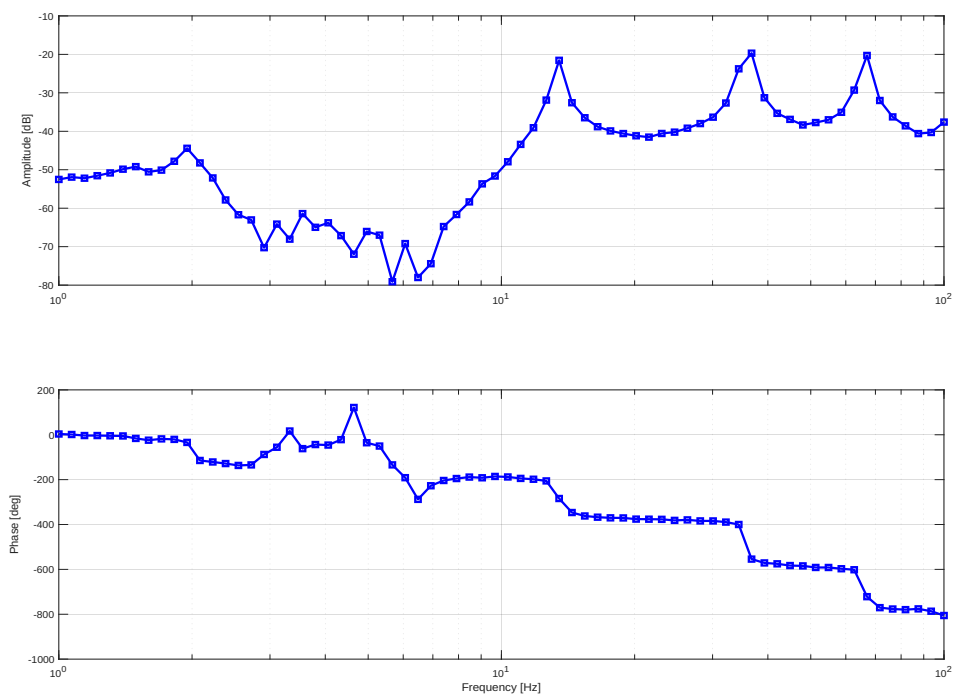


Figure 2.1: Bode plot from 1 Hz to 100 Hz. Sensor at the top of the cantilever.

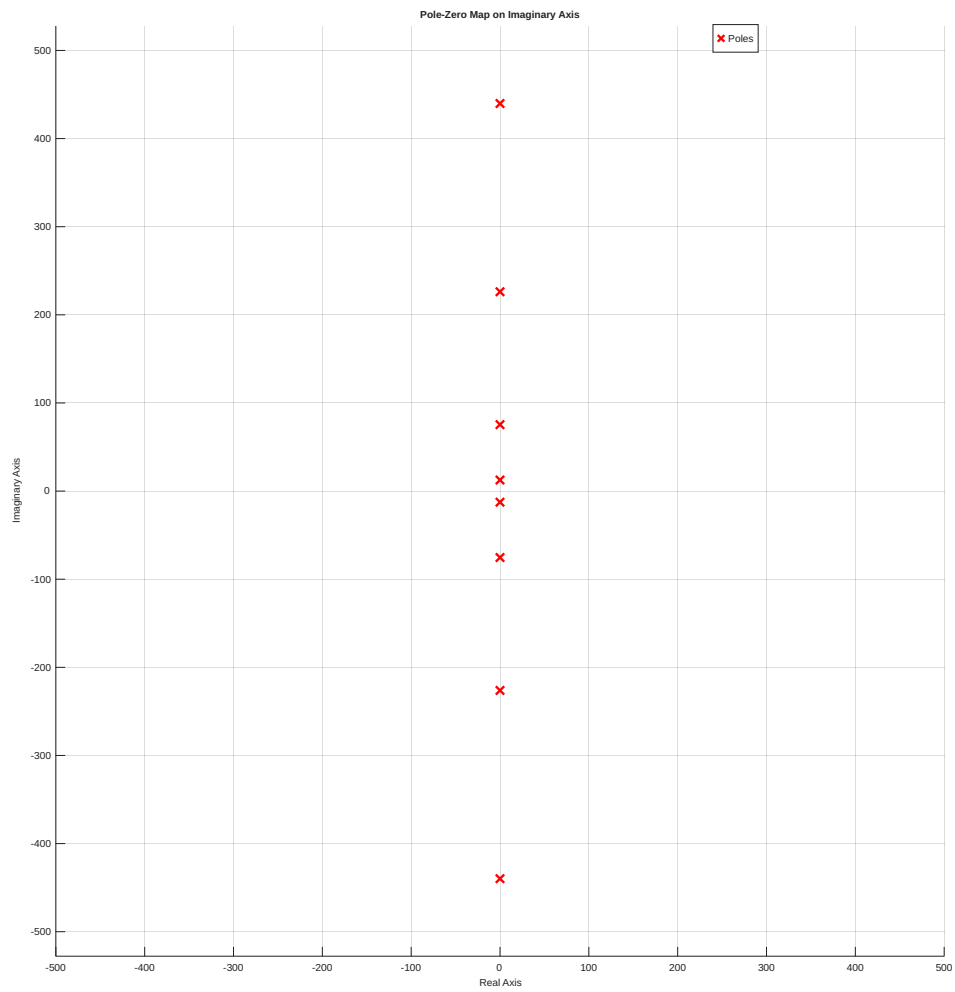


Figure 2.2: Pole-zero map with accelerometer at top

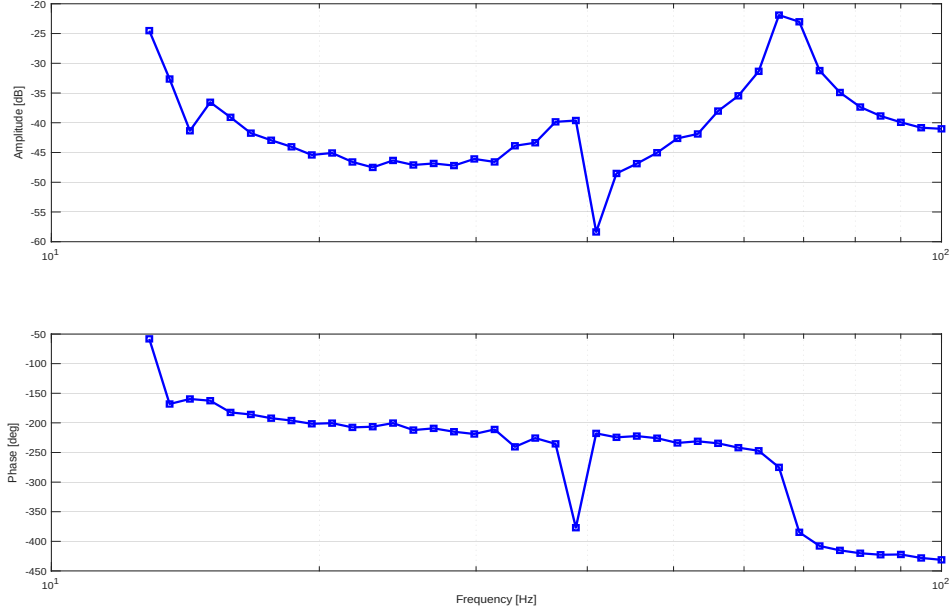


Figure 2.3: Bode plot from 1 Hz to 100 Hz. Sensor placed at the second node of the third mode.

pole at that frequency in the Bode plot)

The measured Bode plot is shown in Figure 2.3. Note that all following Bode plots were measured from 10 Hz to 100 Hz to reduce measurement time. This should be considered when comparing them to the original frequency response in Figure 2.1. Especially because the first mode at about 2 Hz is of course not visible in these plots.

Comparing Figure 2.3 with Figure 2.1, we still observe resonance peaks at approximately 12 Hz and 70 Hz. However, the resonance at approximately 36 Hz, corresponding to the third mode, is not visible. Instead, the magnitude drops sharply at this frequency, indicating a zero in the transfer function. This makes perfect sense, as the sensor is not moving at all for this frequency, which corresponds to very high damping in the magnitude plot. This measurement therefore confirms that the sensor is placed at a node of the third mode, specifically the second node along the beam. The corresponding pole-zero map is given in Figure 2.4.

2.4 System Response — Sensor below a Node

Placing the sensor a couple of centimeters below the previous node, we expect the resonance peak of the third node to be observable in the Bode plot again. The Bode plot is given in Figure 2.5. Our expectation is confirmed,

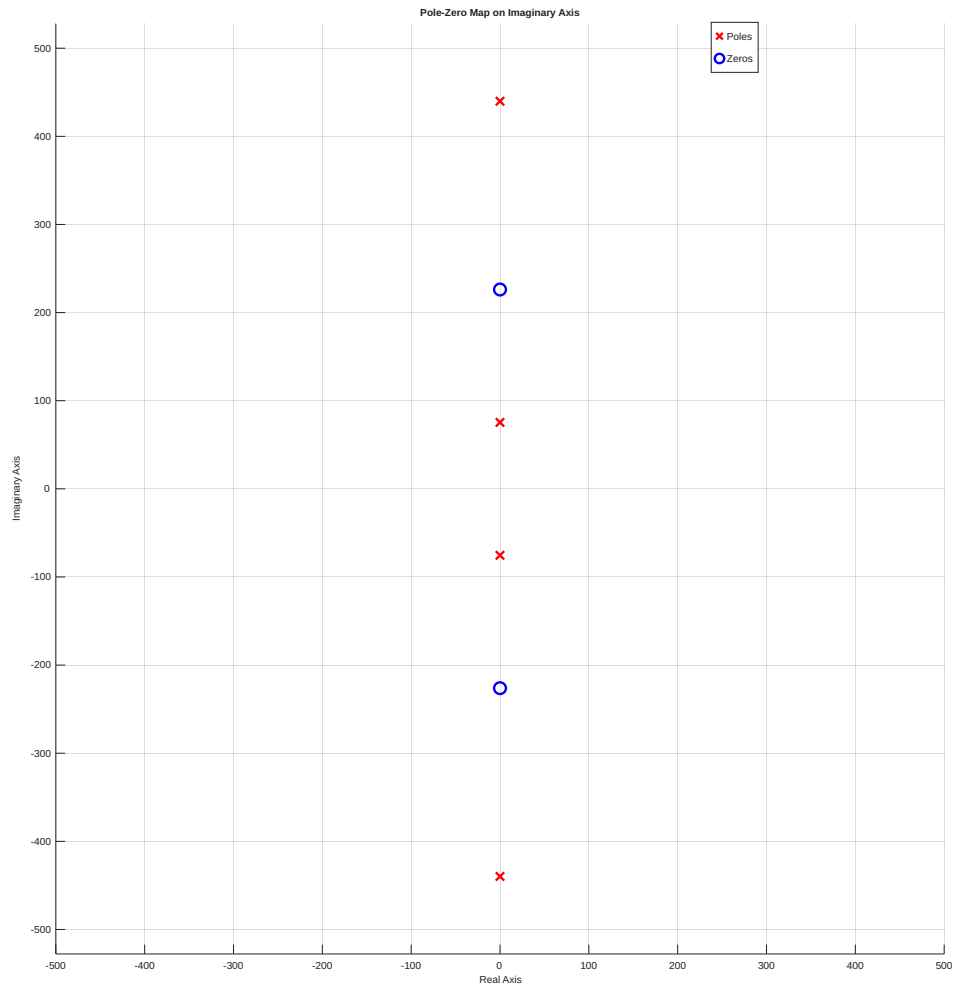


Figure 2.4: Pole-zero map - Sensor placed at the second node of the third mode.

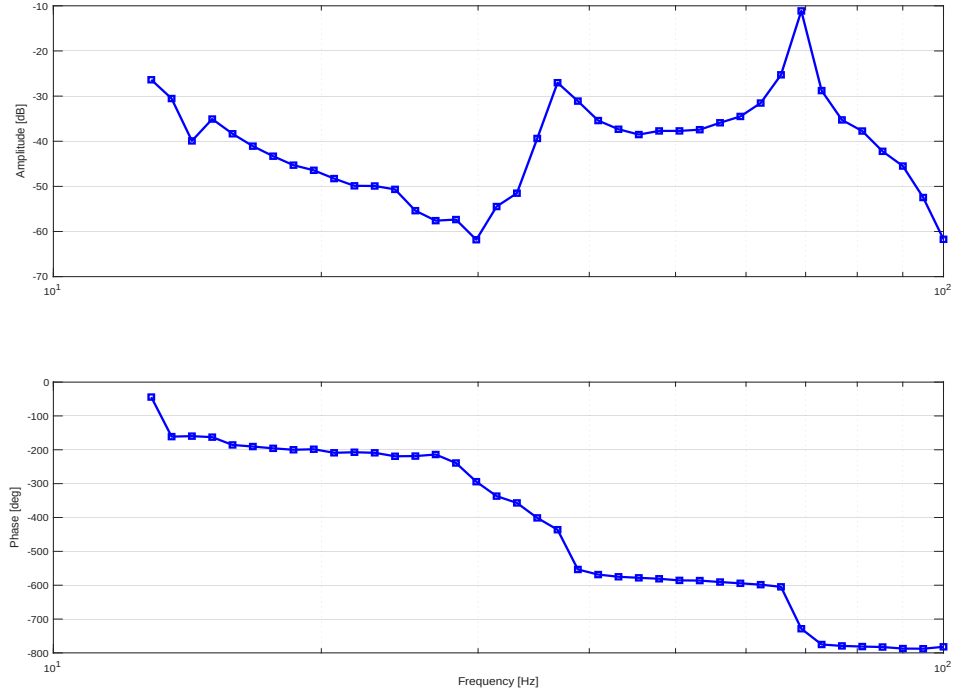


Figure 2.5: Bode plot — Sensor below node of third mode.

as the resonance peak at about 36 Hz is clearly visible in Figure 2.5.

2.5 System Response — Sensor above a Node

Finally, the accelerometer was moved above the node. As expected, we can see the third mode resonance peak again in Figure 2.7. Additionally, we can also observe a zero at about 60 Hz. Because we know there is no resonance at this frequency and the resonance peak at 70 Hz is still clearly visible, we expect this zero to be caused by destructive interference between modes 3 and 4 happening at this specific sensor placement.

2.6 Time Constant and Q-Factor

In this final experiment, the actuator was again positioned a few centimeters above the base of the beam and the accelerometer was placed at the top where the amplitude at the second natural frequency is maximal. We excited the beam at the second resonance frequency ($f_2 = 12$ Hz), then stopped the excitation and measured the decay time until the beam stopped moving. An oscilloscope screenshot of this can be seen in Figure 2.9. From the

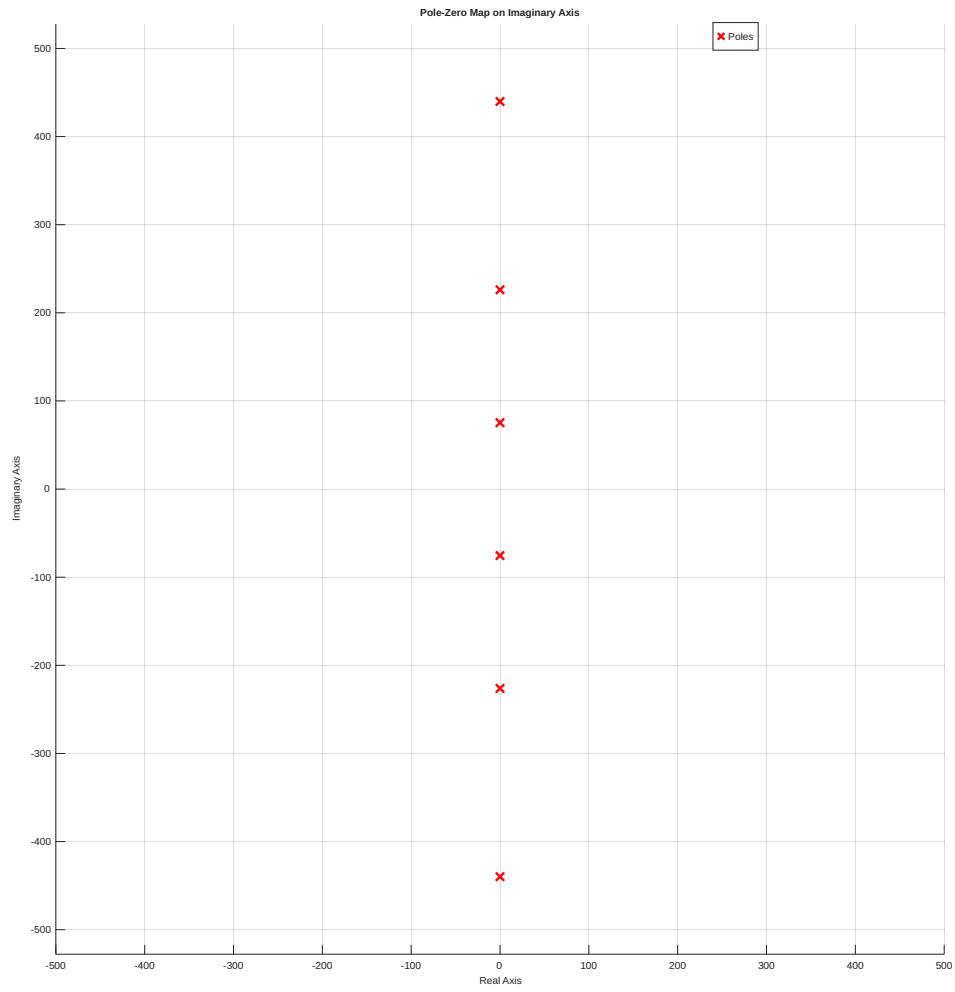


Figure 2.6: Pole-zero map — Sensor below node of third mode.

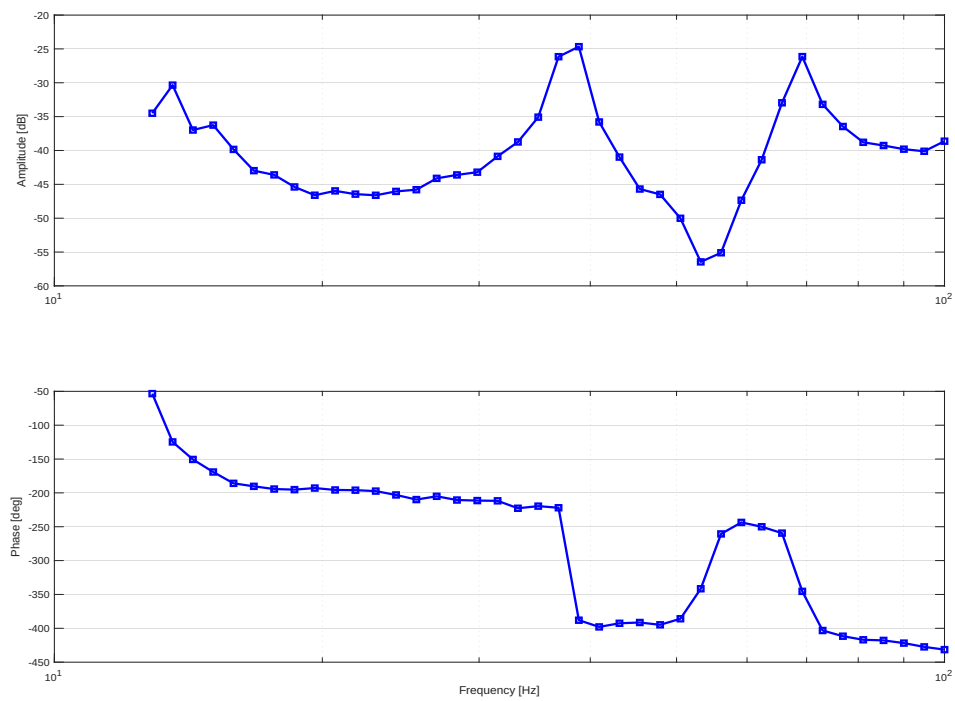


Figure 2.7: Bode plot — Sensor above node of third mode.

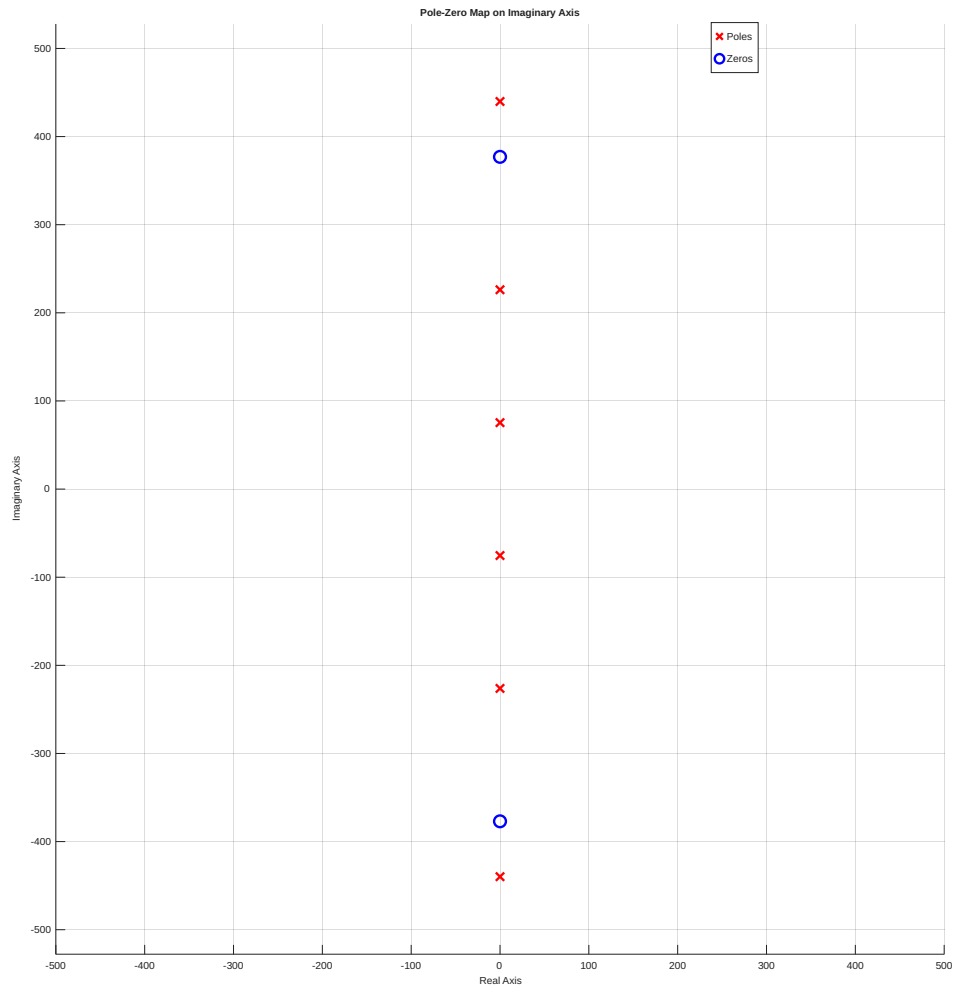


Figure 2.8: Pole-zero map — Sensor above node of third mode.

screenshot we measure the time it takes for the cantilever to return to 5 % of its maximum amplitude:

$$T_s = 1.2 \text{ s.} \quad (2.4)$$

Under the assumption of an underdamped second-order systems, we know the the step response to be a damped oscillation whose envelope decays exponentially. Hence:

$$x(t) \sim e^{-\zeta\omega_n t} \quad (2.5)$$

As we measured the time it takes until our system response decayed to 5 % of the step response maximum, we write

$$e^{-\zeta\omega_n T_s} = 0.05 \quad (2.6)$$

In order to rearrange (2.6) to isolate ζ , we first take the logarithm

$$\zeta\omega_n T_s = -\ln(0.05) \approx 2.996 \quad (2.7)$$

and then solve for ζ :

$$\zeta = \frac{2.996}{\omega_n T_s}. \quad (2.8)$$

As

$$Q = \frac{1}{2\zeta}, \quad (2.9)$$

we can finally calculate

$$Q = \frac{1}{2} \cdot \frac{\omega_n T_s}{2.996} = \frac{\omega_n T_s}{5.99} = \frac{2\pi \cdot 12 \text{ Hz} \cdot 1.2 \text{ s}}{5.99} = 15.1 \quad (2.10)$$

As

$$Q > \frac{1}{2} \quad (2.11)$$

our assumption of an underdamped system is confirmed.

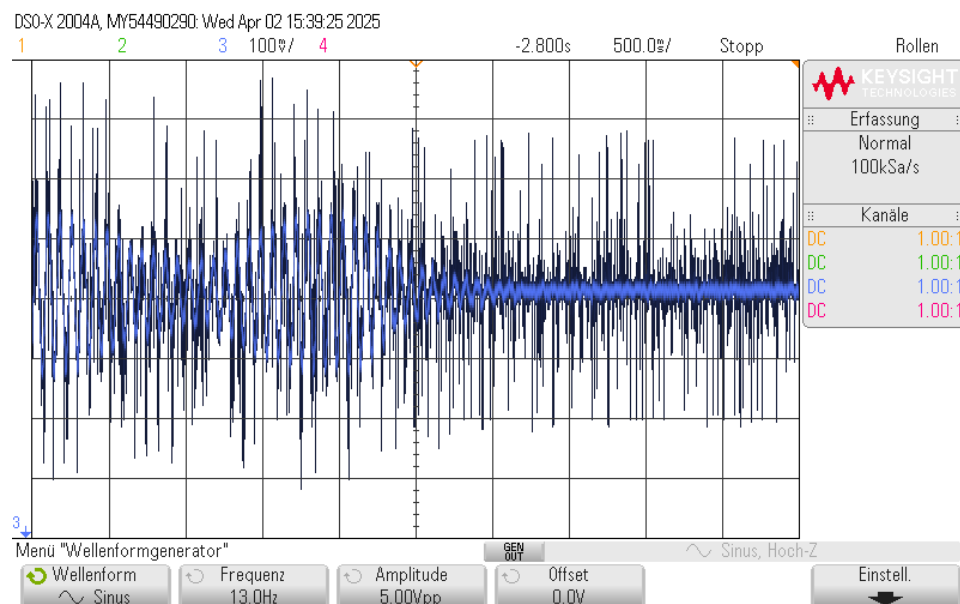


Figure 2.9: Oscilloscope screenshot of the decaying cantilever movement after the excitation is stopped.

Report Questions

3.1 Pole Observability

Depending on the sensor placement, different resonance peaks in the frequency response may or may not be visible. This is perfectly sensible as looking at Figure 1.1, we see the simulated expected cantilever movement, given an excitation of the third resonance frequency, e.g. 36 Hz. This mode shape is of course independent of the sensor placement. However, when placing the sensor directly at a node, e.g. a position along the beam where the mode shape is exactly 0, the beam will not move at this position. Therefore the sensor will measure zero displacement, given excitation of the frequency corresponding to this mode shape. This then yields a zero in the magnitude plot as it is essentially measured as infinite damping. Therefore the expected pole turns into a zero in the Bode plot.

When positioning the sensor above or below the node, the expected resonance peak is visible again in the frequency response.

3.2 Sensor and Actuator Placement

Looking at the second mode shape in Figure 1 we already see the reason for placing the sensor near the top, as this is exactly where the deflection is maximized. As we want to capture the resulting deflection after excitation and the decay of it, it makes sense to place the sensor where this is most visible.

Mounting the actuator near, but not at the bottom, is also sensible. Here, the actuator is far enough away from a node, meaning excitation of the mode is possible. But at the same time, the beam is still relatively stiff near its mounting point at the bottom, allowing for efficient force transfer.

3.3 State-Space Representation

The dynamics of the cantilever beam system are to be approximated as a linear state-space model of the form

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \\ y(t) &= \mathbf{C}\mathbf{x}(t).\end{aligned}\tag{3.1}$$

We model the system such that each mode contributes two states to our system space. That means for an n -mode approximation, the state vector has $2n$ entries

$$\mathbf{x}(t) = [q_1(t) \quad \dot{q}_1(t) \quad q_2(t) \quad \dot{q}_2(t) \quad \cdots \quad q_n(t) \quad \dot{q}_n(t)]^\top. \tag{3.2}$$

Controllability

Controllability can directly be checked with the controllability matrix

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}]. \tag{3.3}$$

The placement of the actuator is represented in the matrix $\mathbf{B}$. For a given mode n , if the mode shape value at the actuator position x_a is zero ($Y_n(x_a) = 0$), then that mode cannot be excited, or in other words, it is not controllable. This is very nice as it perfectly corresponds with our findings in Section 3.2.

Observability

Similarly, we can check observability with the observability matrix

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix}. \tag{3.4}$$

Here, the placement of the sensor directly affects $\mathbf{C}$. If the sensor is located at a node of mode n , i.e. $Y_n(x_s) = 0$, the output is not sensitive to that mode, therefore this mode is not observable.